# Bangkok Transit System (BTS)

**Phakkharada Khongthoeng**
**Panuphat Bannasan**
**Renukarn Yano**

**BECHELOR OF ENGINEERING**
**INFORMATION AND COMPUTER ENGINEERING**

**MAE FAH LUANG UNIVERSITY 2021**

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A COMPUTER ENGINEERING PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
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IN COMPUTER ENGINEERING 2021

EXAMINING COMMITTEE

.........................................................................................ADVISOR
(Asst. Prof. Dr.Suppakarn Chansareewittaya)

.........................................................................................COMMITTEE
(Aj. Dr.Mahamah Sebakor)

......................................................................................... COMMITTEE
(Aj. Ronnachai Sretawat Na Ayutaya)

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Miss Phakkharada Khongthoeng
Mr. Panuphat Bannasan
Miss Renukarn Yano

| | |
|---|---|
| **Title** | Bangkok Transit System (BTS) |
| **Author** | Miss Phakkharada Khongthoeng |
| | Mr. Panuphat Bannasan |
| | Miss Renukarn Yano |
| **Degree** | Bachelor of Engineering (Computer Engineering) |
| **Supervisory Committee** | Asst. Prof. Dr.Suppakarn Chansareewittaya Advisor |
| | Aj. Dr.Mahamah Sebakor Committee |
| | Aj. Ronnachai Sretawat Na Ayutaya Committee |

# Abstract

Experience the convenience of online ticket purchase for the Bangkok Skytrain on our user-friendly platform. Navigate seamlessly through the bustling city with our streamlined ticketing system, allowing you to purchase and manage your Skytrain tickets from the comfort of your device. Explore the various routes, check real-time schedules, and enjoy a hassle-free commute with secure and efficient online transactions. Embrace the future of urban transit with our online ticketing service, making your journey through Bangkok's dynamic landscape both effortless and enjoyable.
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# CONTENTS

# LIS OF TABLE

# LIST OF FIGURES

# CHAPTER 1
# INTRODUCTION

## 1.1 Background and Rational

In Thailand, rapid economic growth and population mobility have been prominent in recent years. Bangkok, along with its neighboring areas, stands as the hub of the country's progress. Investments in urban development and amenities have strengthened the quality of life in the city, resulting in an increased desire for urban migration.

The mass transit systems in Bangkok, namely the BTS and MRT, play a pivotal role in promoting efficient and convenient mobility for both residents and tourists. These systems encourage a shift towards the use of public transportation, offering convenience and a unique experience for passengers. Traditionally, purchasing train tickets in Thailand has involved long queues and the risk of ticket loss or damage. Recognizing the need for a healthier and more efficient travel environment, we aim to create an online ticketing website, making public transportation more accessible.

The introduction of online ticketing not only enhances convenience but also contributes to the sustainability and development goals of the country. Thailand is committed to continually improving the lives of its citizens, prioritizing modern technology and user convenience in shaping the future of transportation and public transit in the nation.

## 1.2 Objective

1.2.1 Enhance Passenger Convenience

1.2.2 Promote Public Transportation Usage

1.2.3 Improve Travel Efficiency

1.2.4 To study, learn, and develop a web application system

## 1.3 Scope

1.3.1 Registration as a member is a prerequisite for accessing the system

1.3.2 Capable of accessing information regarding the routes and stations of BTS and MRT within a city

1.3.3 Users have the capability to submit reports for any issues encountered within the system to our attention

1.3.4 Can make reservations, receive tickets, and make online payments through

the platform

## 1.4 Methodology

1.4.1 Proposal preparation and defense

1.4.2 Literature review and requirement gathering

1.4.3 Design website and system design

1.4.4 Front-end development and database implementation

1.4.5 Senior project 1 document preparation

1.4.6 Senior Project 1 exam

1.4.7 Testing website

1.4.8 Senior Project 2 exam document preparation

1.4.9 Senior Project 2 exam

## 1.5 Plan

**Table 0.1** Working plan.

| | Pre-project | | | | |
|---|---|---|---|---|---|
| **Plan/Month** | Jan | Feb | Mar | Apr | May |
| Proposal preparation and defense | | | | | |
| Literature review and requirement gathering | | | | | |
| Design website and system design | | | | | |
| Front-end development and database implementation | | | | | |
| Senior Project 1 presentation | | | | | |

| | Proj ect | | | | | | |
|---|---|---|---|---|---|---|---|
| **Plan/Month** | Feb | Mar | Apr | May | Jun | July | Aug |
| Front-end development and database implementation | | | | | | | |
| Testing website | | | | | | | |
| Senior Project 2 presentation | | | | | | | |

## 1.6 Expected Result

1.6.1 Developed Next.js and JavaScript coding skills and created a prototype website related to online railway ticket

## 1.7 Place and Equipment

### 1.7.1 Place

1.7.1.1 Mae Fah Luang University
1.7.1.2 Dormitory

### 1.7.2 Software

1.7.2.1 Visual Studio Code

1.7.2.2 Xampp

1.7.2.3 Figma

### 1.7.3 Language and tool development

1.7.3.1 Next.js

1.7.3.2 JavaScript

1.7.3.3 Node.js

# CHAPTER 2
# LITERATURE REVIEW

## 2.1 Related Research

The creation of an online railway ticket purchasing website in Thailand is of significant importance due to several compelling reasons. Firstly, it enhances passenger convenience as the online system allows people to book and purchase tickets with ease, eliminating the need to spend time queuing at departure stations, reducing rush-hour congestion. This serves as an attractive option and supports the future development of public transportation systems. Moreover, efficient data management and real-time status checks for railway services can be achieved through this platform, ensuring efficient transportation and time savings for passengers. Ultimately, the development of such a website also presents opportunities for digital businesses and contributes to the advancement of information technology in Thailand, offering both value and feasibility.

## 2.2 Related Work

**1. Design website and system design**

In this section, we have meticulously planned and designed the UX/UI system to ensure that users can easily and efficiently purchase tickets. Our website design is divided into four main parts, each with a well-defined purpose:

1. Homepage and Features: This section serves as the starting point for users and includes essential information about the train service, ticket search functionality, and links to the main sections of the website.

2. Maps and Station Information: Here, users can access the system's map and find detailed information about each train station, including departure times. It provides users with the ability to view routes and select specific stations.

3. Online Ticket Booking: The final section allows users to swiftly and efficiently book train tickets online. Users can select their preferred route, date, time, and the number of passengers.

This system and website design aim to make the ticket purchasing process convenient and efficient for users while providing a highly satisfying and engaging user experience.

**2. Front-end development and database implementation**

Front-end development is about creating the user interface and experience, while database implementation handles data storage and retrieval. Front-end involves UI/UX design, programming languages, responsive design, interactivity, and cross-browser compatibility.

Database implementation includes design, choosing a DBMS, data modeling, storage, retrieval, security, scalability, and optimization. Together, they create functional and user-friendly software and websites.

In summary, front-end development focuses on crafting the user interface and experience, while database implementation is responsible for managing the crucial data that the application relies on. These two components work in harmony to deliver functional, user-friendly software applications, and websites that meet both design and data needs.

**3. Testing website**

Website testing is a crucial phase in web development, ensuring that a website meets high standards before going live. It involves systematic evaluation of various aspects, including functionality, compatibility, responsive design, performance, security, usability, content accuracy, cross-device and cross-browser compatibility, regression, load handling, accessibility, A/B testing, database integrity, caching, and the use of browser developer tools. Effective testing guarantees a smooth user experience, preventing potential issues that could harm the website's reputation or security. It is an ongoing process as websites evolve and change.

## 2.3 Related Technology

### 2.3.1 Software

#### 1. Visual Studio

Visual Studio is a powerful developer tool that you can use to complete the entire development cycle in one place. It is a comprehensive integrated development environment (IDE) that you can use to write, edit, debug, and build code, and then deploy your app. Beyond code editing and debugging, Visual Studio includes compilers, code completion tools, source control, extensions, and many more features to enhance every stage of the software development process.

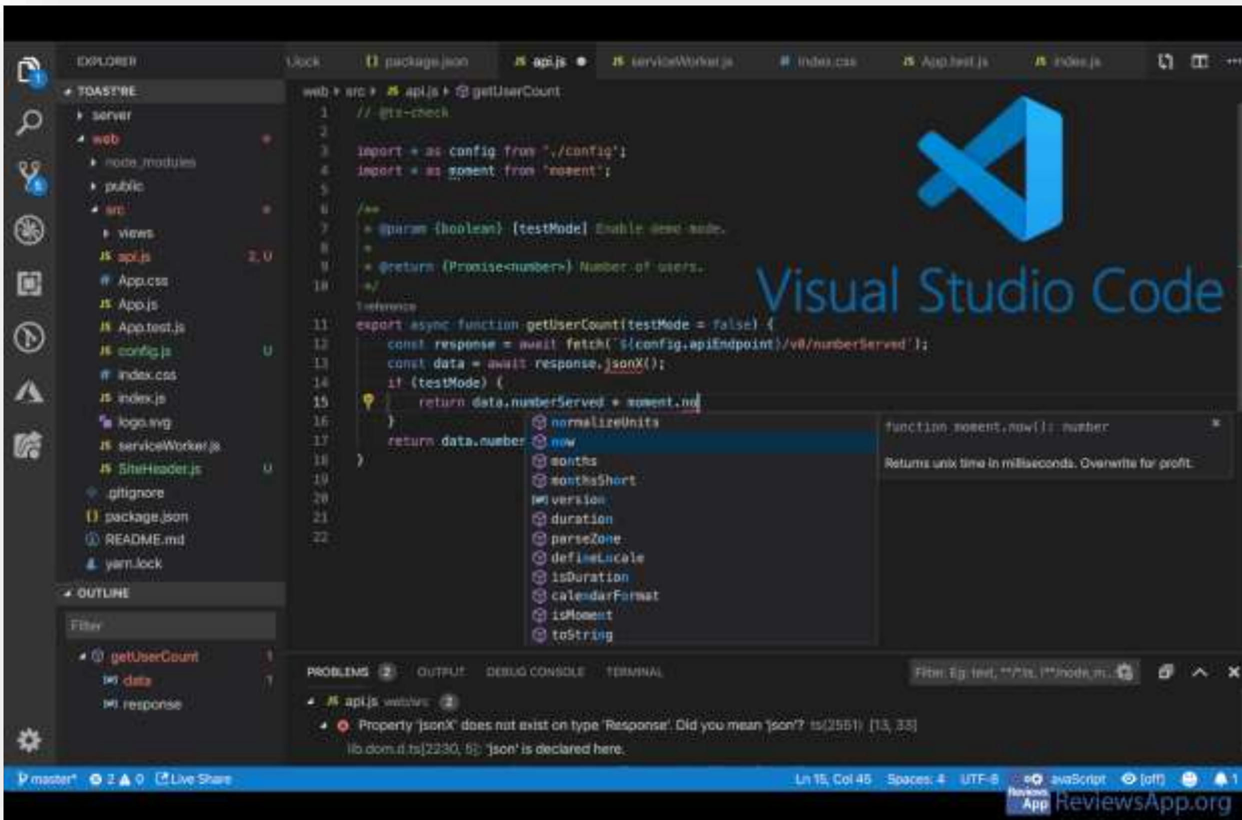


**Figure 2.1:** Visual Studio Code window

## 2. XAMPP

XAMPP is a free and open-source software package that provides a local development environment for web developers. The name XAMPP is an acronym that stands for:

1. X: Stands for "cross-platform," meaning it can run on various operating systems like Windows, Linux, macOS, and more.
2. A: Stands for "Apache," which is a popular web server software.
3. M: Stands for "MySQL," which is a widely used relational database management system (RDBMS).
4. P: Stands for "PHP," which is a scripting language commonly used for web development.
5. P: Stands for "Perl," which is another scripting language, although it is less commonly used in web development compared to PHP.

In essence, XAMPP is a bundled software stack that includes Apache, MySQL, PHP, and Perl. It also often includes other tools and components like phpMyAdmin (a web-based MySQL administration tool) and FileZilla (an FTP server). Developers use XAMPP to create a local web server environment on their own computers, which allows them to develop and test websites or web applications before deploying them to a live server on the internet. This local development environment simulates the conditions of a real web server, making it easier for developers to work on their projects locally.

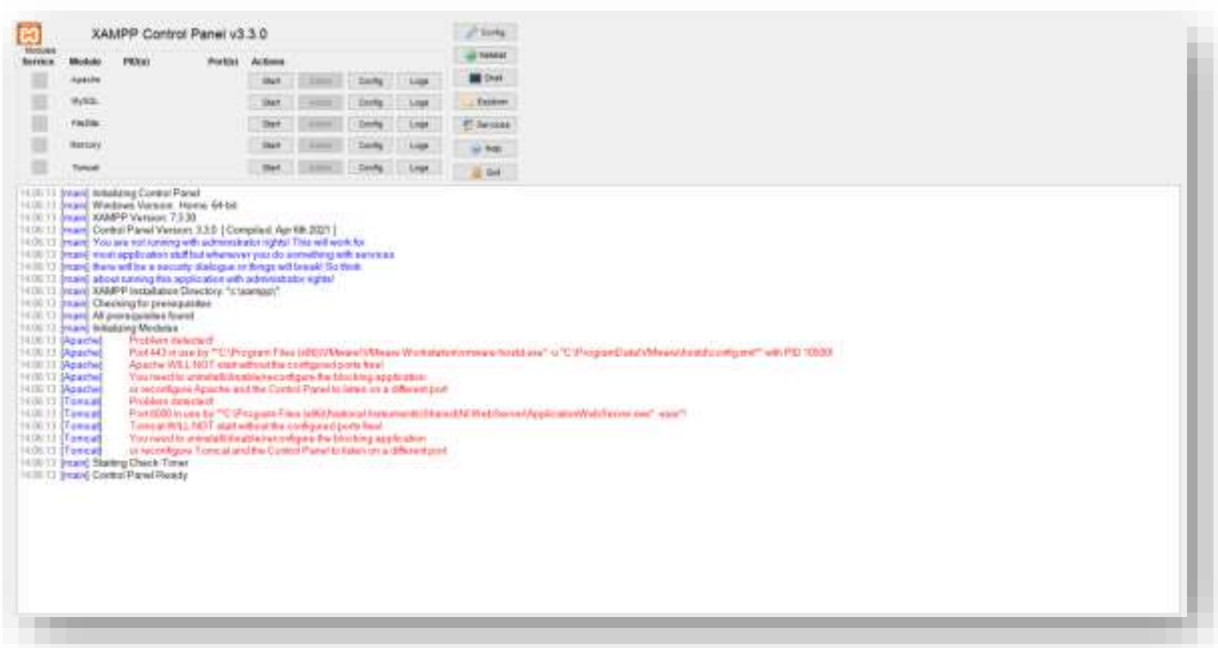


**Figure 2.2:** XAMPP on window

### 3. Figma

Figma is a vector graphics editor and prototyping tool primarily web-based, with additional offline features accessible through desktop applications for macOS and Windows. The Figma mobile app for Android and iOS enables the viewing and interaction with Figma prototypes in real-time on mobile devices. Focused on user interface and user experience design, Figma's feature set emphasizes real-time collaboration.

Positioned as a design platform for collaborative product development, Figma is tailored for teams. Originating on the web, Figma facilitates the creation, sharing, testing, and finalization of designs for teams building products together. Whether streamlining tools, simplifying workflows, or fostering collaboration across teams and time zones, Figma accelerates the design process, enhancing efficiency and enjoyment while ensuring everyone stays synchronized.

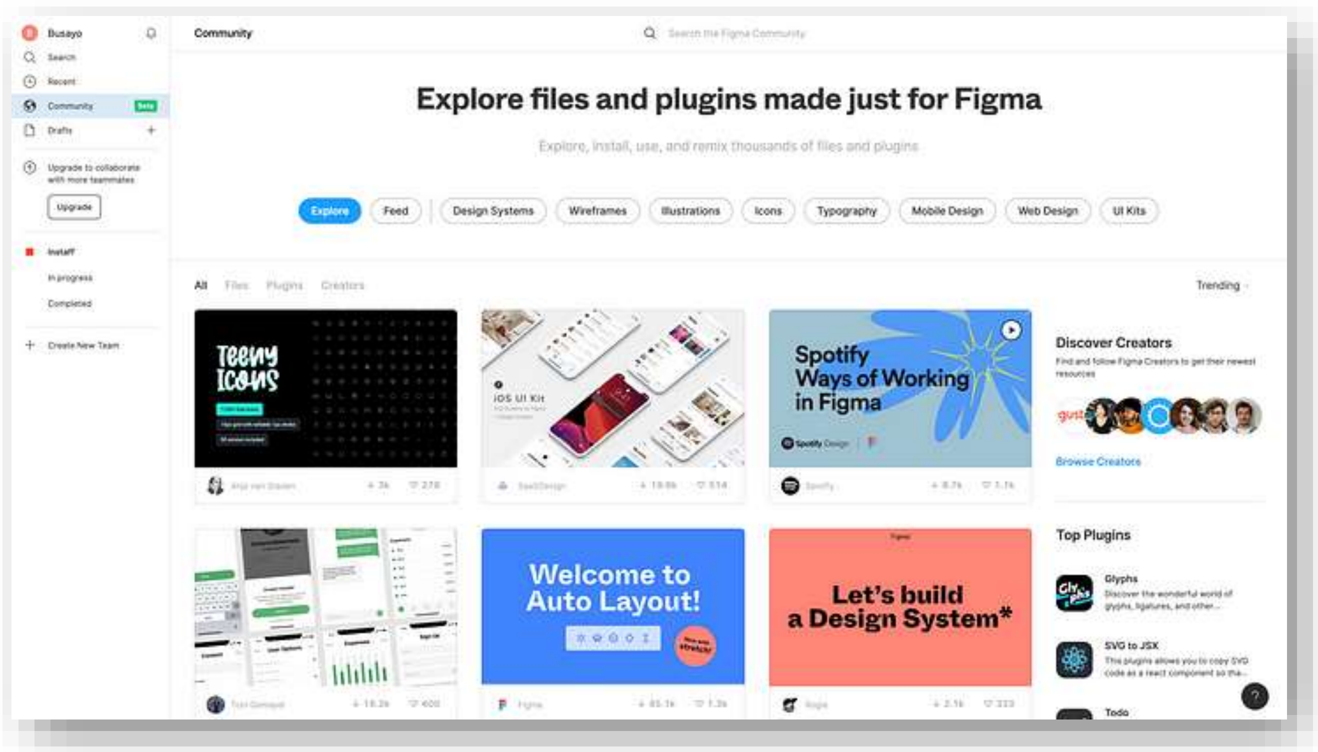


**Figure 2.3:** Example of UI design wit Figma

### 2.3.2 Language and tool development

#### 1. Next.js

Next.js is an open-source React-based web framework designed for streamlined web application development. It simplifies React applications by offering conventions and utilities. Key features include server-side rendering, static site generation, automatic code splitting, intuitive routing, API route creation, robust CSS support, and extensibility through plugins. Widely used in modern web development, Next.js is favored for its flexibility, performance optimization, and developer-friendly approach, particularly in the React community.

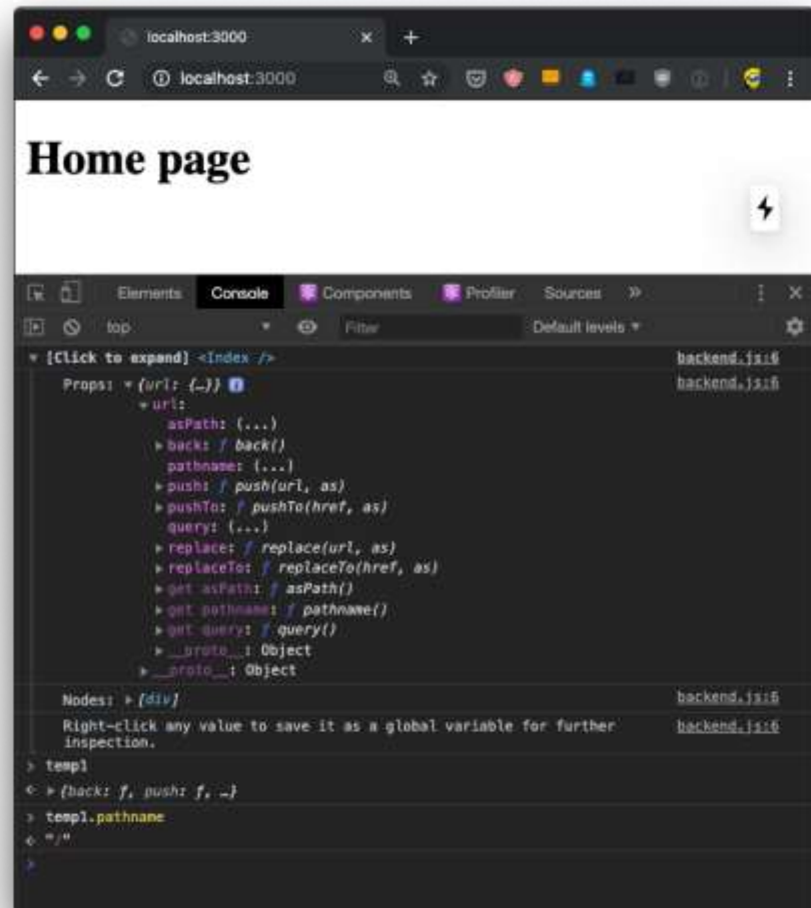


**Figure 2.4:** Example of Next.js code

**2. JavaScript**

JavaScript is a versatile high-level programming language widely recognized for its pivotal role in web development. Used alongside HTML and CSS, it empowers developers to craft dynamic and interactive web content. Key features include client-side scripting for dynamic user interfaces, object-oriented capabilities, event-driven programming, support for asynchronous operations, cross-browser compatibility, and its utilization in both frontend and backend development through platforms like Node.js. JavaScript is foundational in web development, driving the creation of responsive and engaging user experiences.


**Figure 2.5:** Example of JavaScrip code

### 3. Node.js

Node.js is an open-source, cross-platform runtime environment for back-end JavaScript. It operates on the V8 engine, executing JavaScript code outside of a web browser. Node.js enables the use of JavaScript for command line tools and server-side scripting, unifying web application development under a 'JavaScript everywhere' paradigm. With an event-driven architecture and support for asynchronous I/O, Node.js optimizes throughput and scalability, making it suitable for web applications with numerous I/O operations and real-time functionality.

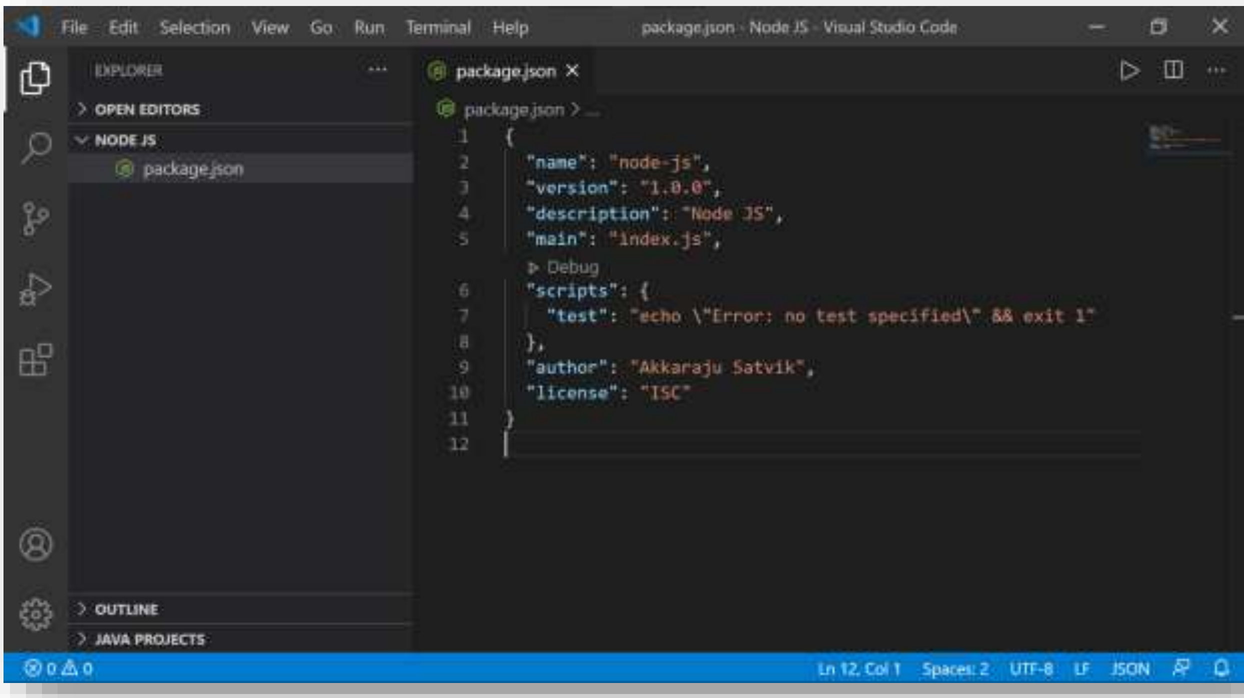


**Figure 2.6:** Example of Node.js code

# CHAPTER 3
# METHODOLOGY

Methodology for Bangkok Transit System (BTS). Defined Methodology in 3 steps as follows.

## 3.1 target group

The target group for individuals wishing to travel within Bangkok.

## 3.2 System Development

### 3.2.1 Problem Definition

During the coding process within Visual Studio, developers may encounter various challenges. Visual Studio, recognized as a robust integrated development environment (IDE) extensively used for software development across multiple programming languages, presents a spectrum of functionalities and tools. However, within this expansive toolkit, our team encountered a specific challenge that necessitated careful attention and problem-solving. This issue serves as a testament to the dynamic nature of coding environments and underscores the importance of adapting strategies to address unique hurdles encountered during the development lifecycle.

**1. Configuration and Setup Problems:**

When configuring Visual Studio for a specific project or language, developers may encounter various difficulties, leading to potential configuration errors or issues throughout the project setup process. Establishing the right configuration is crucial for a smooth development workflow, and overcoming these challenges often requires a careful understanding of the project's requirements, programming language specifications, and the intricacies of Visual Studio's configuration settings. Effectively addressing these configuration and setup issues is fundamental to ensuring a stable and efficient development environment.

**2. Compilation Errors:**

During the development process, code may encounter compilation failures, often arising from syntax errors, missing dependencies, or other issues. These compilation errors can present significant challenges in the subsequent debugging phase. Identifying and resolving these issues require meticulous attention to detail and a comprehensive understanding of the programming language, project dependencies, and the intricacies of the compilation process. Successfully navigating through these compilation errors is essential for ensuring a robust and error-free codebase, ultimately contributing to the efficiency and success of the development endeavor.

**3. Debugging Challenges:**

Navigating the process of identifying and rectifying bugs in code can prove to be intricate, and developers may encounter challenges in effectively leveraging Visual Studio's debugging tools. Debugging is a critical phase in the software development lifecycle, requiring a keen eye for detail and a deep understanding of the codebase. Visual Studio offers a comprehensive suite of debugging features, but the effective utilization of these tools demands familiarity and expertise. Developers may need to employ various debugging techniques to trace and resolve issues.

**4. Integration Issues:**

Given Visual Studio's frequent integration with a diverse range of frameworks, libraries, and plugins, developers may encounter compatibility problems or integration issues, potentially impeding the seamless progression of the development workflow. Ensuring smooth collaboration between different components is crucial for the success of a project, and resolving integration challenges requires a nuanced understanding of the involved technologies. Developers may need to conduct thorough compatibility checks, address versioning conflicts, and implement effective troubleshooting strategies to optimize the integration process and maintain a cohesive development environment.

**5. Performance Concerns:**

In the realm of software development, undertaking resource-intensive tasks or working on large-scale projects within Visual Studio can potentially give rise to performance issues, adversely affecting the overall development experience. As projects grow in complexity and scale, developers may encounter challenges related to response times, memory utilization, and overall system responsiveness. Mitigating these performance concerns necessitates a strategic approach, including optimizing code, configuring Visual Studio settings, and employing profiling tools to identify and address bottlenecks. Proactively managing performance issues contributes significantly to maintaining a smooth and efficient development environment.

**6. Version Control Problems:**

The integration of Visual Studio with version control systems, particularly popular ones like Git, can give rise to a spectrum of challenges. These challenges may manifest as conflicts, merge issues, or difficulties in effectively managing code repositories. Collaborative development efforts often involve multiple contributors working simultaneously on codebases, and without proper version control management, discrepancies may arise. Resolving version control problems requires a nuanced understanding of branching strategies, conflict resolution techniques, and effective communication among team members. Successfully navigating these challenges ensures a streamlined and coordinated approach to version control, contributing to the overall success of collaborative software development projects.

### 3.3.2 Analysis

We analyze how to design diagrams and other detail of work and we design 2 diagrams is Use case Diagram and Class Diagram.

3.3.2.1 Use case Diagram

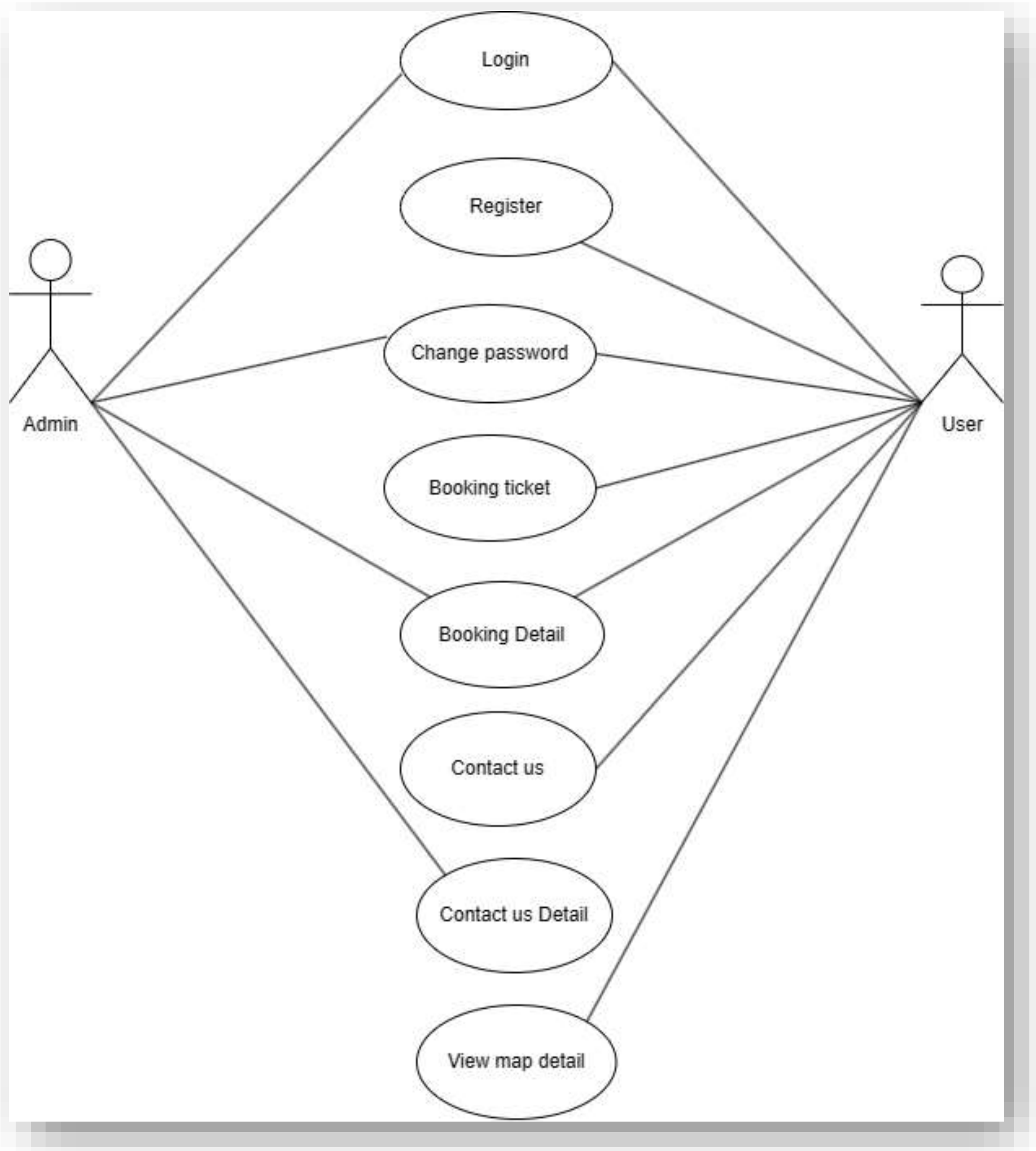


**Figure 3.1:** Use case Diagram

3.3.2.1 Class Diagram

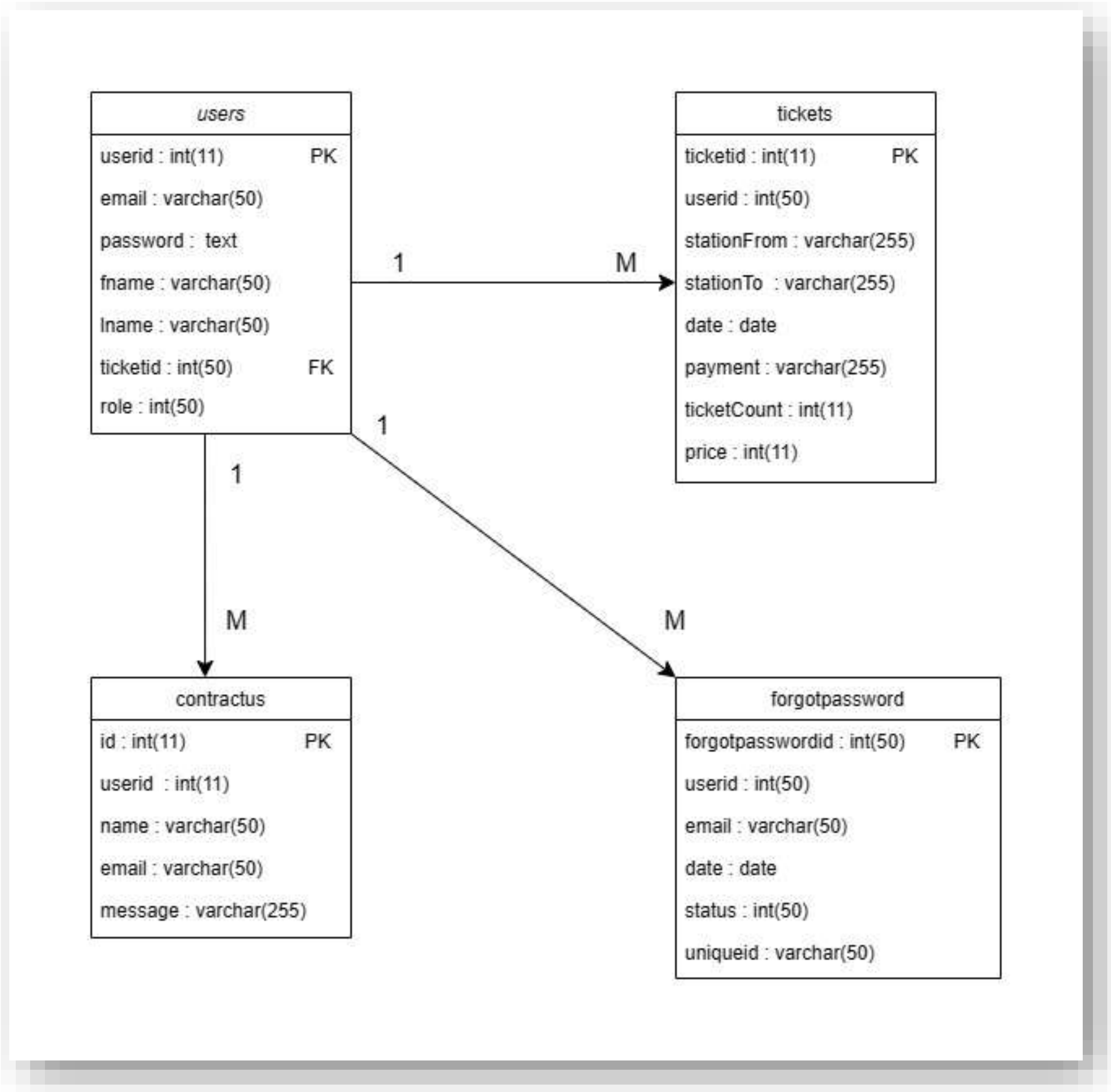


**Figure 3.2:** Class Diagram

## 3.3 Design

The design of website are very important for user when they are using our website, we might design it for user friendly and modern design.

1. **Register page**

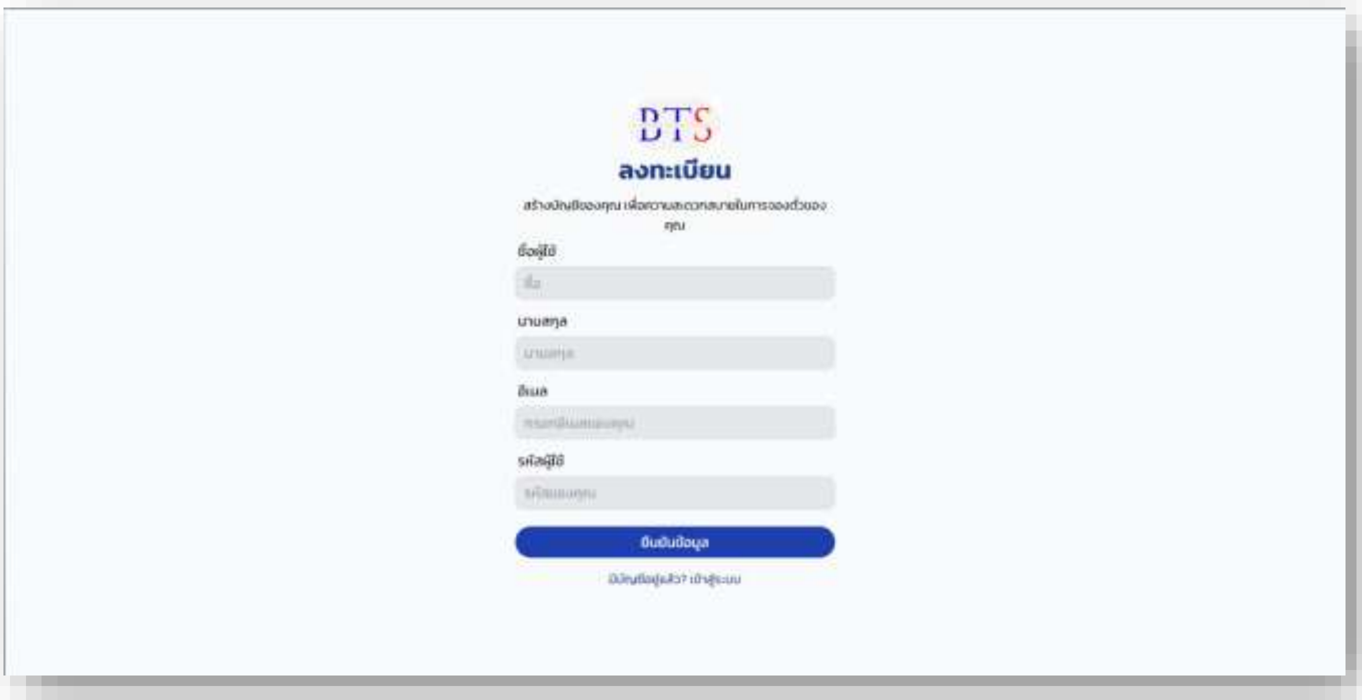


**Figure 3.3:** Register page

Figure 3.3: Registration page, or the page designed for users to register, plays a pivotal role in the establishment of user accounts on a website, serving as a fundamental step in the onboarding process

**2. Log-in page**

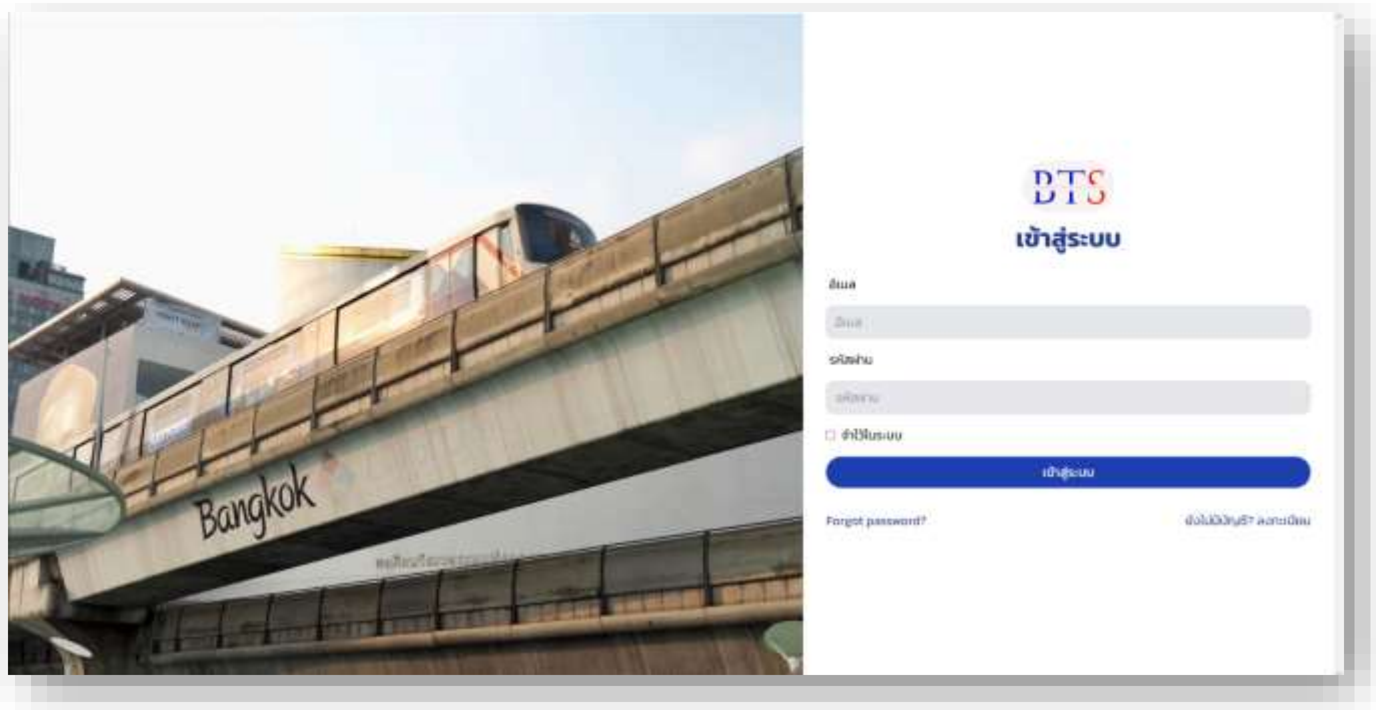


**Figure 3.4:** Log-in page

Figure 3.4: Log-in page is vital for users to access their accounts, requiring authentication information like username and password for various features and services, including personal information

## 3. Homepage

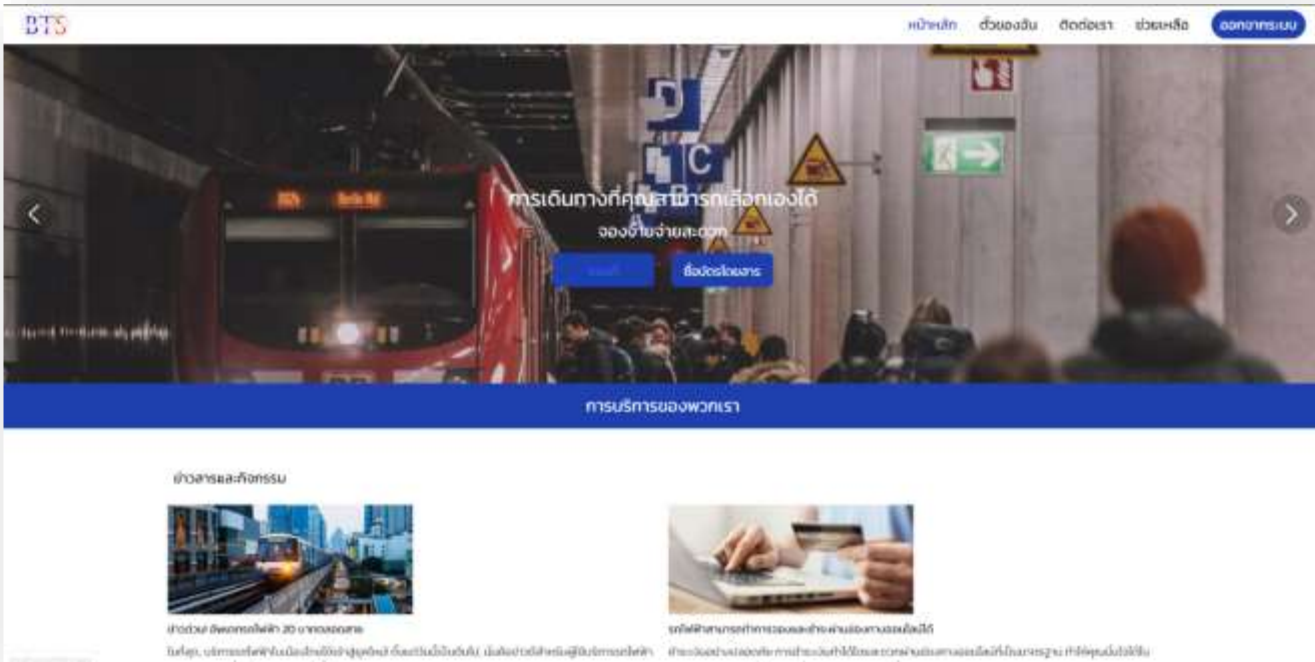


**Figure 3.5:** Homepage

Figure 3.5: Homepage, or main page, serves as the initial interface users encounter when accessing a website or application. It features two primary functionalities: ticket reservation and viewing the subway map. The top section of the webpage includes a menu for various accesses such as My Booking, Contact Us, and Help Center. The bottom section provides relevant news and information related to the website

## 4. Map metro page

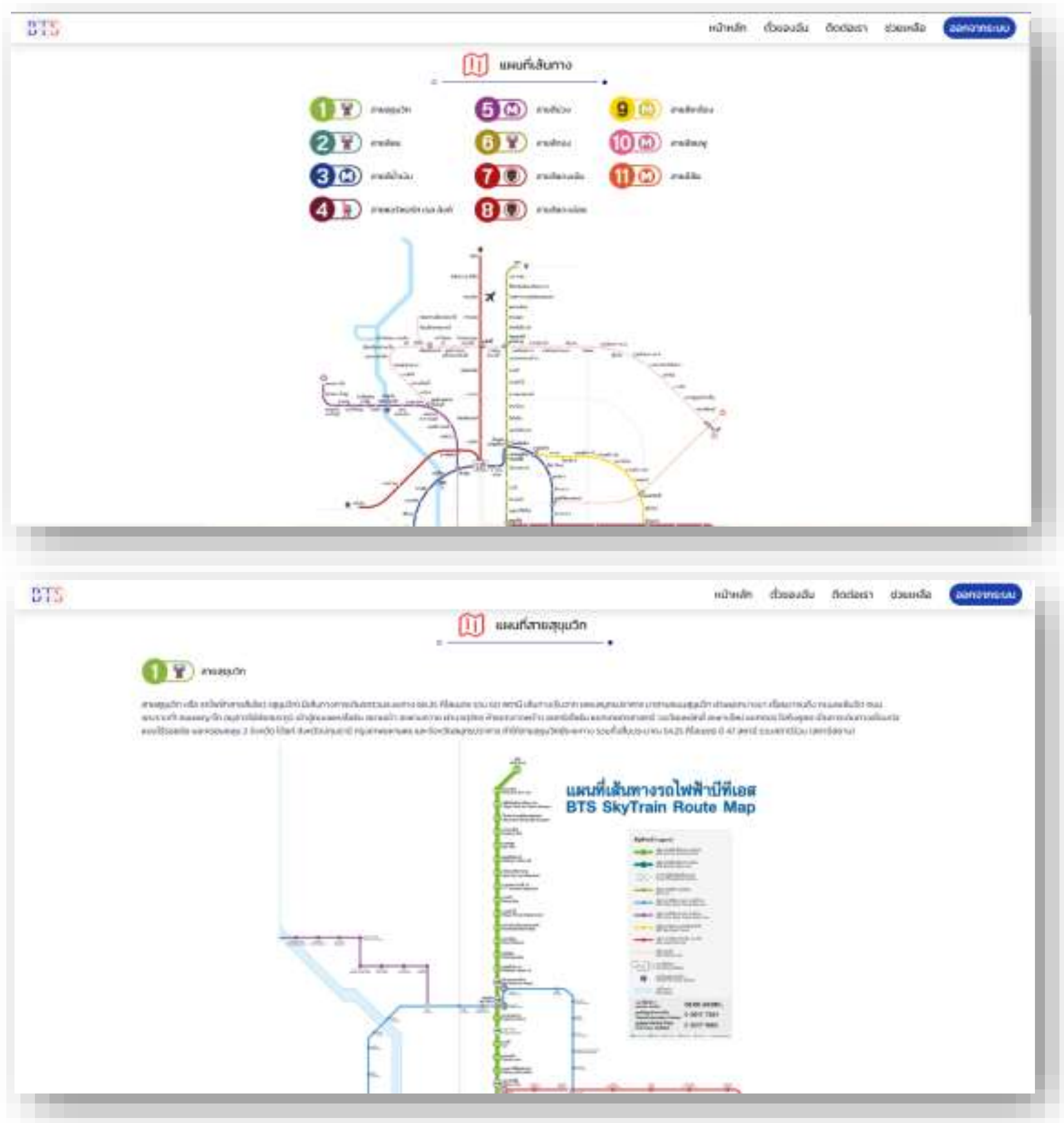


**Figure 3.6:** Map metro page

Figure 3.6: Metro Map Page is a web page that provides information about the routes and stations of the metro or public transportation system in a city. This page is designed to help users easily understand and navigate to various locations within the city

### 5. Booking page

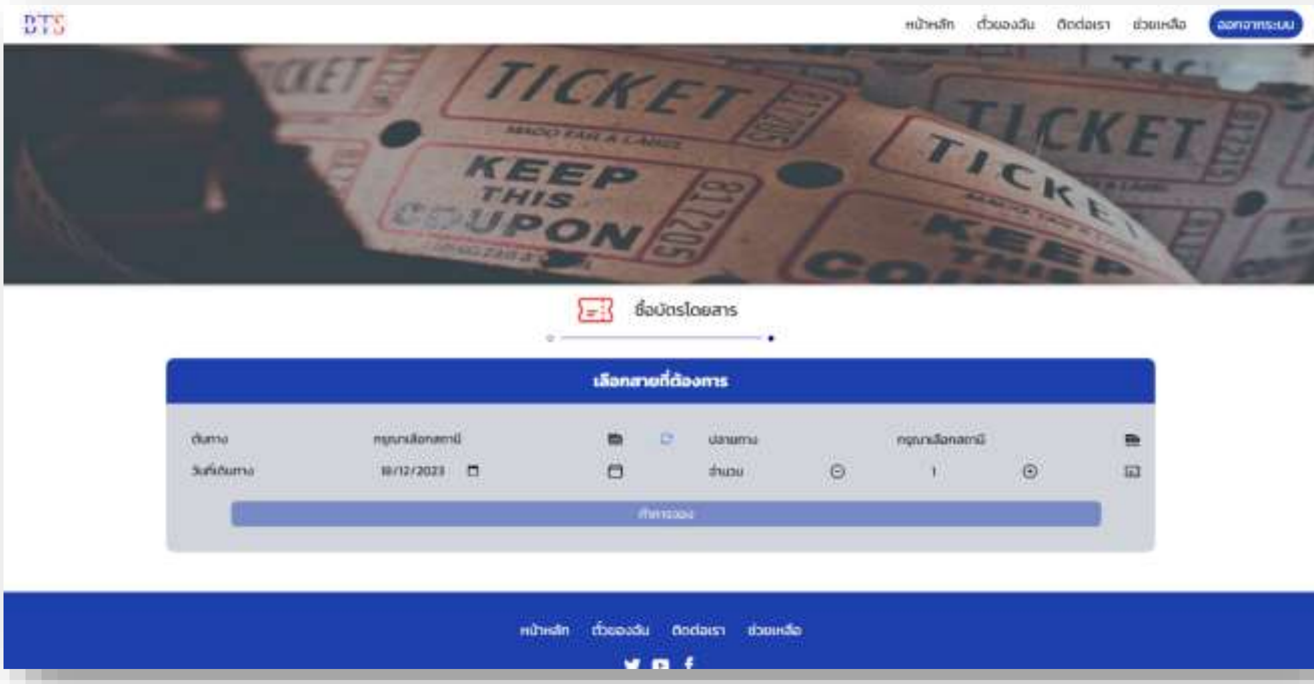


**Figure 3.7:** Booking page 1


**Figure 3.8:** Booking page 2

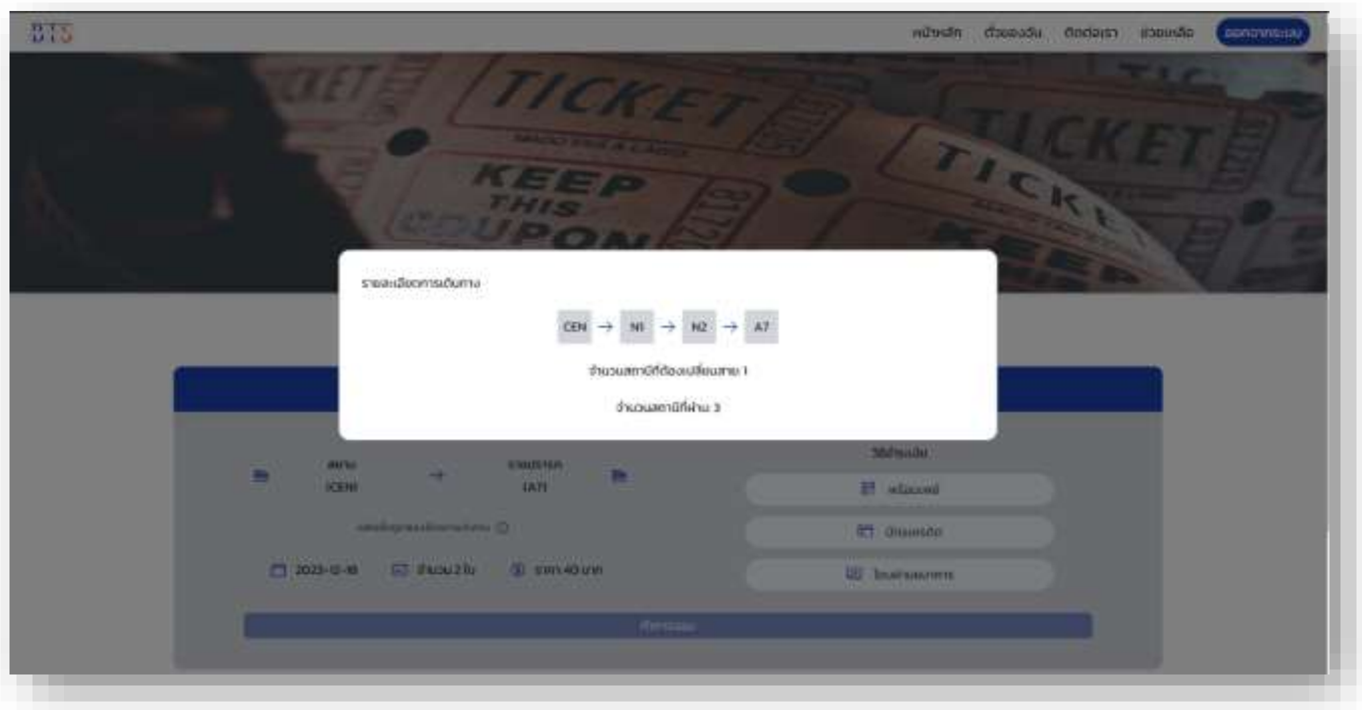


**Figure 3.9:** Booking page 3

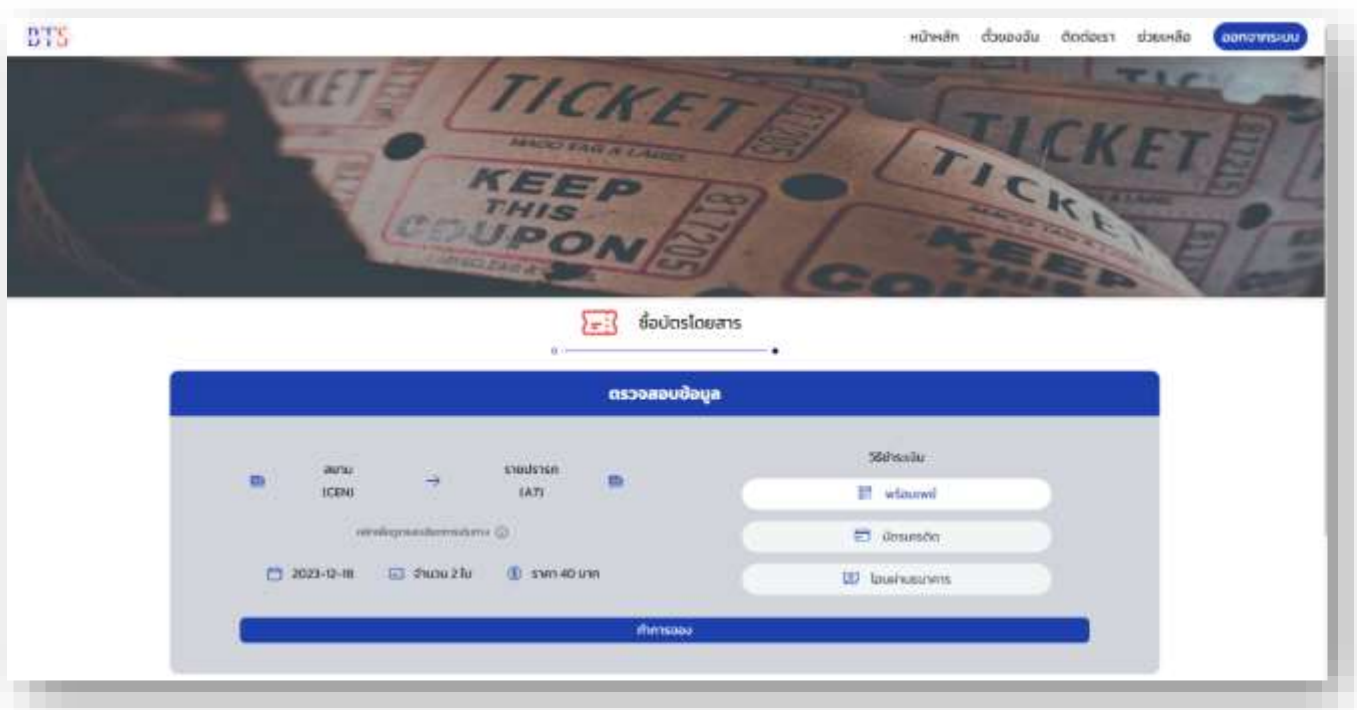


**Figure 3.10:** Booking page 4

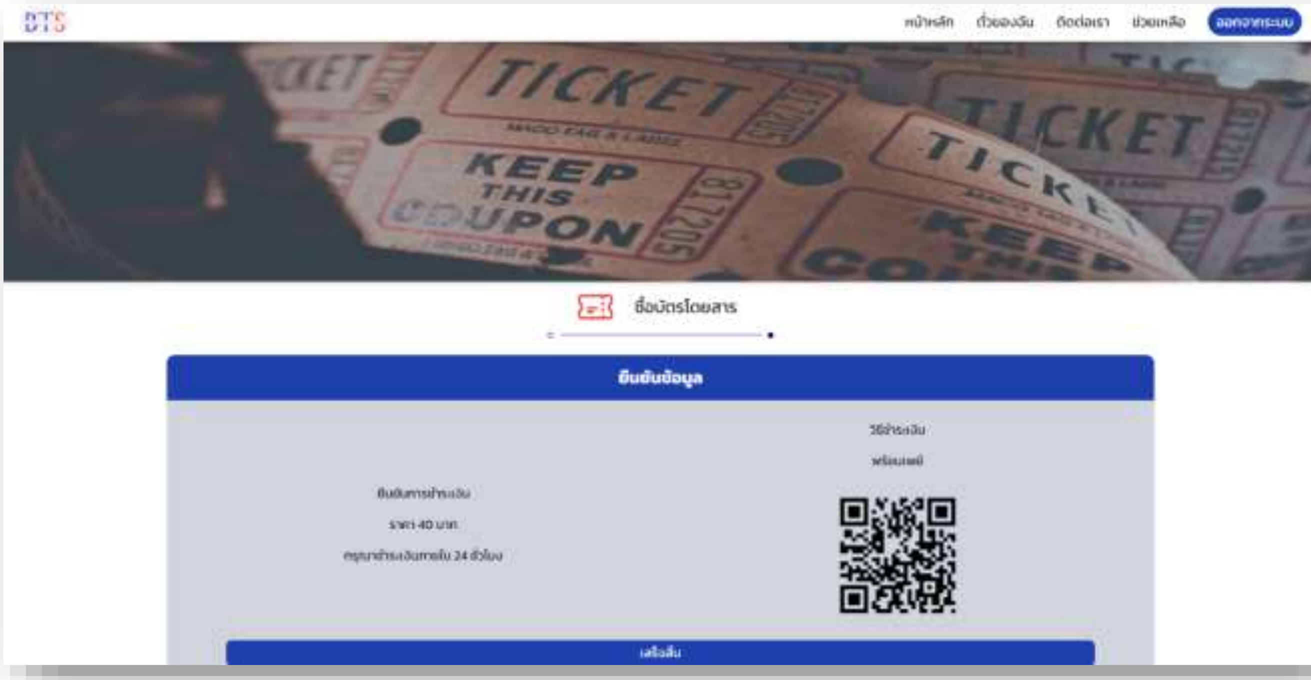


**Figure 3.11:** Booking page 5

Figure 3.7 – 3.11**:** Booking Page, or Reservation Page, is a dedicated section of the website enabling users to initiate online reservations for services or procure tickets. Its design focuses on enhancing the convenience and efficiency of the booking process for users

6. My booking page

**Figure 3.12:** My booking page

Figure 3.12: My Booking page is a component of the website that allows users to view and manage previously made reservations. This page typically includes features that enable users to review all their booking information, view details, and perform various management actions related to the reservations

## 7. Contract us page

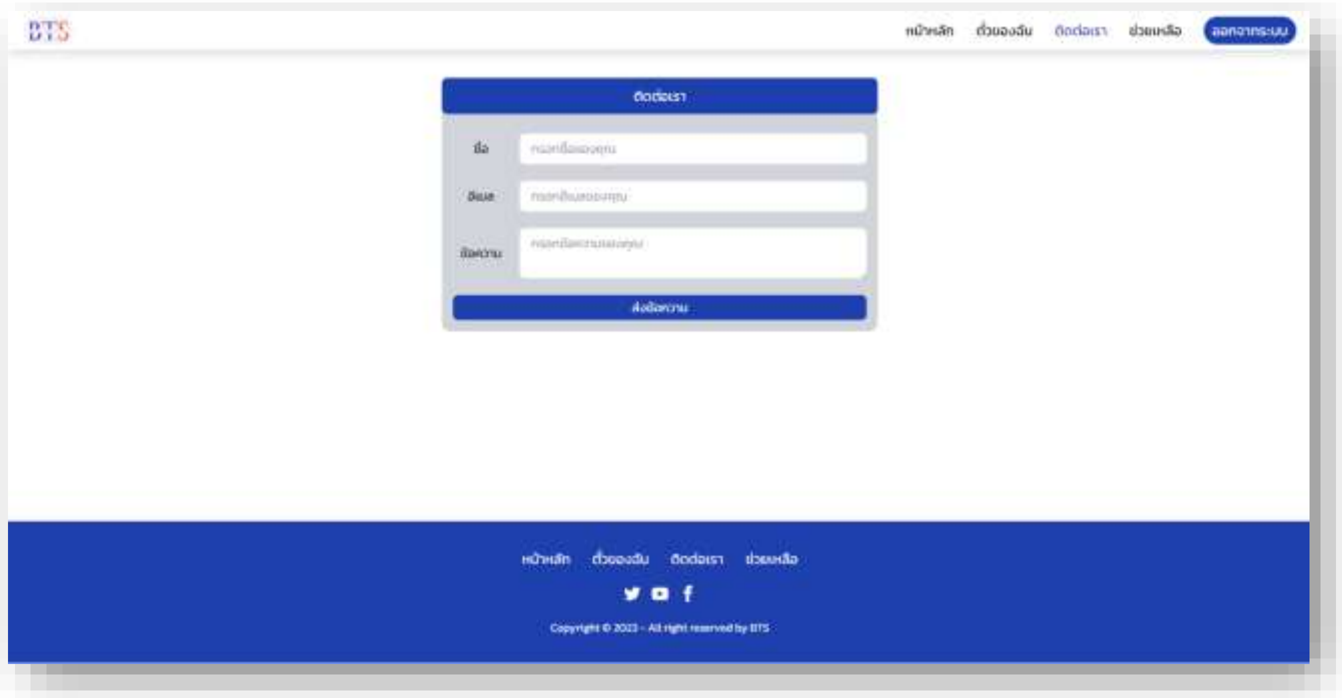


**Figure 3.13:** Contract us page

Figure 3.13: Contact Us page is an integral section of our website that enables users to communicate with the site administrators or customer service team. This page is designed to allow users to send messages, ask questions, or make other forms of contact with the customer service team

## 8. Help center page

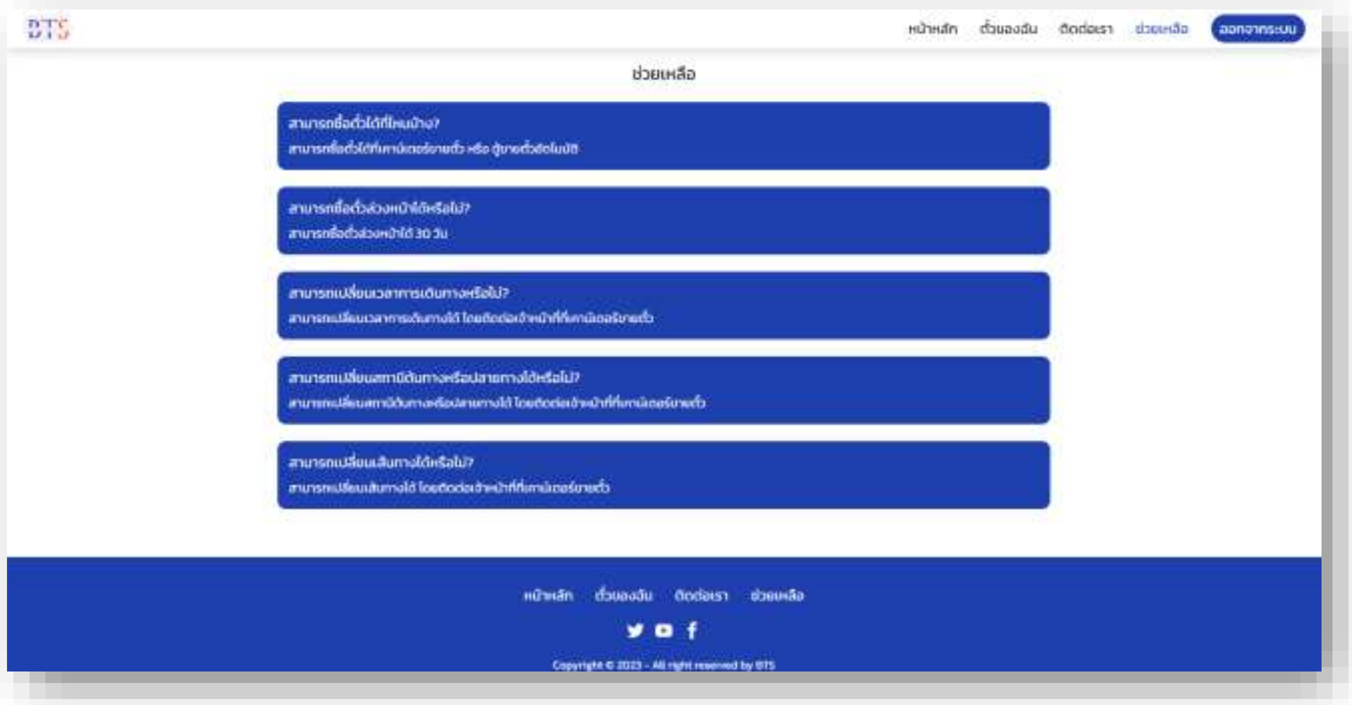


**Figure 3.14:** Help center page

Figure 3.14: Help Center page is an integral component of our website, providing information and resources to assist users in resolving issues, offering guidance, or providing additional information for any questions or concerns they may have. This page is typically designed to be a contemporary and user-friendly source of information for our users.

## 9. Admin page

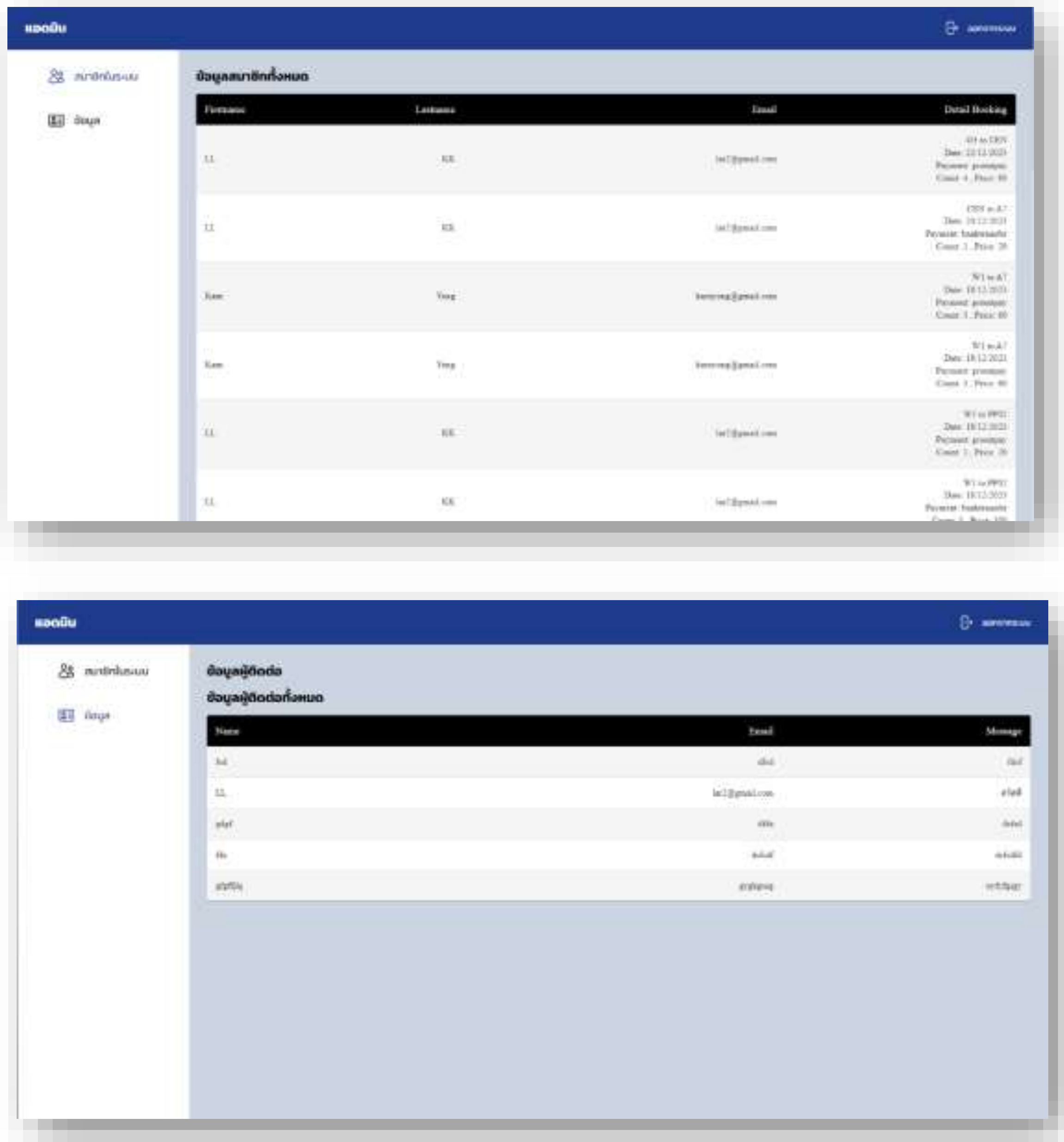


**Figure 3.15:** Help center page

Figure 3.15: Admin Page is a crucial section of the website that grants privileges to system administrators (Admin) for managing and controlling various functions of the system. This page is equipped with high-level encryption and security measures to ensure that only those with the highest privileges can access and perform operations on this page.

# CHAPTER 4

# CONCLUSION

## 4.1 Conclusion

In conclusion, our online ticketing platform for the Bangkok Skytrain redefines the way you experience urban transit in the city. With seamless online ticket purchase and management, navigating through Bangkok becomes a hassle-free and efficient journey. Embrace the future of digital convenience as you explore routes, check real-time schedules, and securely purchase your Skytrain tickets from the comfort of your device. Elevate your commuting experience with our user-friendly platform, making every trip through the vibrant cityscape of Bangkok a convenient and enjoyable adventure. Welcome to the future of online transit, where your journey begins with just a click.

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